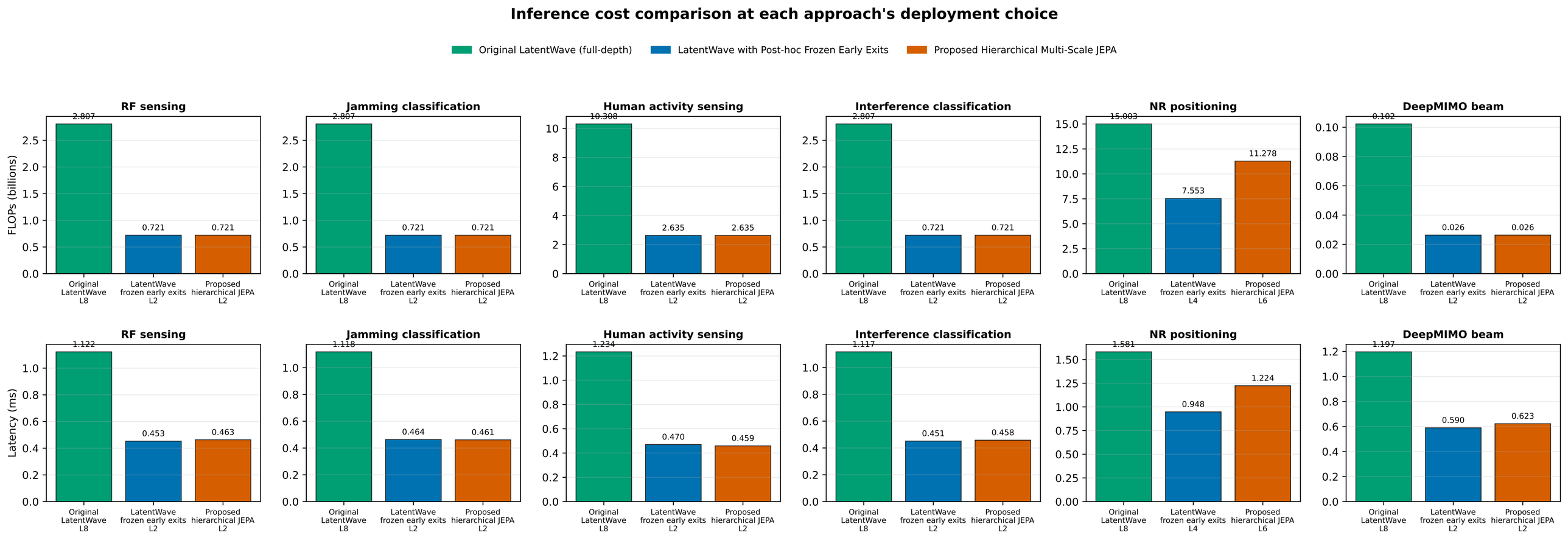
Inference cost comparison at each approach's deployment choice
Original LatentWave (full-depth)
LatentWave with Post-hoc Frozen Early Exits
Proposed Hierarchical Multi-Scale JEPA
RF sensing
Jamming classification
Human activity sensing
Interference classification
NR positioning
DeepMIMO beam
FLOPs (billions)
Latency (ms)
2.807
0.721
0.721
2.807
0.721
0.721
10.308
2.635
2.635
2.807
0.721
0.721
15.003
7.553
11.278
0.102
0.026
0.026
1.122
0.453
0.463
1.118
0.464
0.461
1.234
0.470
0.459
1.117
0.451
0.458
1.581
0.948
1.224
1.197
0.590
0.623
Original LatentWave L8
LatentWave frozen early exits L2
Proposed hierarchical JEPA L2
LatentWave frozen early exits L4
Proposed hierarchical JEPA L6
Batch-1 synchronized CUDA timing on the RTX 5070 Ti; lower is better. Layer labels are printed below each method.